

Subject Description Form

Subject Code	AAE3003
Subject Title	Aircraft Propulsion Systems
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	To provide students with knowledge of advanced modern aircraft gas turbine engines
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Possess a deep theoretical understanding of state-of-the-art propulsion systems, including underlying mechanical engineering principles and relevant aircraft gas turbine engines; - Effectively apply knowledge, skills, and hands-on experience to investigate and analyze the key properties of propulsion systems in various engineering contexts. - Function professionally in multidisciplinary teams (including the knowledge of aviation engineering to different situation of engineering context and professional practices in propulsions systems).
Subject Synopsis/ Indicative Syllabus	Introduction to propulsion – Brief history of propulsion, aircraft piston engine, propeller, turbojet engine, aircraft gas turbine engines (turbojet, turbofan, turboprop, turboshaft), performance parameters of turbofan engines, other air-breathing engines, rockets Thermodynamics – System and its properties, measurement diagnostics, first law of thermodynamics, forms of work and heat interactions, ideal gas law, internal energy, enthalpy, specific heat capacities, mass conservation and momentum conservation for steady flow process, the second law of thermodynamics, entropy, heat engines, gas power cycles (Carnot cycle, Otto cycle, Diesel cycle, Brayton cycle). Compressible flows – Speed of sound, Mach number, stagnation properties, critical properties, nozzle flow, shock waves, measurement diagnostics and flow visualization Aircraft engine subsystems – Nacelle, inlet, nozzle, fan, compressor, turbine, combustor, afterburner, fluid system (air, oil, and fuel), control system, engine test and maintenance, engine start. Engine performance parameters, parametric cycle analysis, and component performance - turbojet, turbojet with afterburner, turbofan, turboprop, and turbo-shaft engines.

Teaching/Learning Methodology	1. The teaching and learning methods include lectures, homework assignments, lab reports, and a final examination. 2. The continuous assessment and examination are aimed at providing students with the integrated knowledge required for propulsion systems. 3. Technical/practical examples and problems are raised and discussed in class. 4. Experiments projects are designed to evaluate the propulsion system. <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="3">Intended subject learning outcomes to be covered</th></tr><tr><th>a</th><th>b</th><th>c</th></tr><tr><td>1. Lectures</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Homework assignments</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>3. Experiments/Projects</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>4. Final exam</td><td>✓</td><td>✓</td><td></td></tr></table>	Teaching/Learning Methodology	Intended subject learning outcomes to be covered			a	b	c	1. Lectures	✓	✓	✓	2. Homework assignments	✓	✓	✓	3. Experiments/Projects	✓	✓	✓	4. Final exam	✓	✓						
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Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="3">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th></tr><tr><td>1. Lab session</td><td>30%</td><td></td><td>✓</td><td>✓</td></tr><tr><td>2. Homework assignments</td><td>20%</td><td>✓</td><td></td><td>✓</td></tr><tr><td>3. Examination</td><td>50%</td><td>✓</td><td>✓</td><td></td></tr><tr><td>Total</td><td>100%</td><td colspan="3"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 0.5 × End of Subject Examination + 0.5 × Continuous Assessment The continuous assessment consists of homework assignments and lab reports. They are aimed at evaluating the progress of students' study, assisting them in self-monitoring of fulfilling the respective subject learning outcomes and enhancing the integration of the knowledge learned. The examination is used to assess the knowledge acquired by the students for understanding and analyzing the problems critically and independently, as well as to determine the degree of achieving the subject learning outcomes.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			a	b	c	1. Lab session	30%		✓	✓	2. Homework assignments	20%	✓		✓	3. Examination	50%	✓	✓		Total	100%			
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Student Study Effort Expected	Class contact:	
	▪ Lecture	30 Hrs.
	▪ Lab	6 Hrs.
	Other student study effort:	
	▪ Week 7 reading week	3 Hrs.
	▪ Self-study	39 Hrs.
	▪ Lab preparation and homework assignment	39 Hrs.
	Total student study effort	117 Hrs.
Reading List and References	1. Thermodynamics: An Engineering Approach, 8th Edition, 2014, by Yunus A. Cengel and Michael A. Boles. McGraw-Hill Education 2. Fluid Mechanics: Fundamentals and Applications, 4th Edition, 2018. Cengel, Y. & Cimbala, J., McGraw-Hill Education 3. Elements of Propulsion: Gas Turbine and Rockets, 2nd Edition, 2016. Mattingly, J. AIAA. 4. The Jet Engine, 5th Edition, 2015, Rolls Royce, Wiley 5. Aircraft Propulsion and Gas Turbine Engines, 2nd Edition, 2017, El-Sayed, A. F.	